

Installing Chromaprint (`fpcalc`) on Windows

The acoustic (sound-based) fingerprinting features of this project need a small free tool called `fpcalc.exe`, which is part of **Chromaprint**. This guide walks you through installing it, step by step. No programming experience needed.

In a hurry? Just run the automatic installer instead:

open PowerShell in this folder and run

```
powershell -ExecutionPolicy Bypass -File .\install_chromaprint.ps1
```

(see [Option A](#) below). The manual steps are

here in case you prefer to do it yourself.

What is `fpcalc` and why do I need it?

- `fpcalc.exe` reads audio and produces an **acoustic fingerprint** (a digital “signature” of how something sounds).
- The Python package `pyacoustid` (already in `requirements.txt`) calls `fpcalc.exe` behind the scenes.
- Without `fpcalc.exe`, the phonetic (dialogue) features still work, but the acoustic / hybrid features (`--acoustic`, `--both`) will fail with an error like “fpcalc not found”.

You only need to do this **once**.

Two ways to install

Option	Best for	Effort
A. Automatic (PowerShell or <code>.bat</code>)	Most people	Easiest — one command
B. Manual	If scripts are blocked, or you like doing it yourself	A few clicks

Option A: Automatic (recommended)

This project ships with two ready-made installers. Pick whichever you prefer.

A1. PowerShell installer (most capable)

1. Open the project folder in **File Explorer** (the folder that contains this file, `INSTALL_WINDOWS.md`).

2. Click the address bar at the top, type `powershell`, and press **Enter**.

A blue PowerShell window opens **already in this folder**.

3. Paste this command and press **Enter**:

```
powershell
```

```
powershell -ExecutionPolicy Bypass -File .\install_chromaprint.ps1
```

1. The script downloads Chromaprint, extracts `fpalc.exe`, and asks whether you want to:

- **(1)** copy `fpalc.exe` into this project folder (simplest), or
- **(2)** add it to your user **PATH** (so every program can find it).

Type **1** or **2** and press **Enter**.

5. When it finishes it runs `fpalc -version` to prove it works. Done!

If you ever see a red “running scripts is disabled on this system” message, use the exact command above — the `-ExecutionPolicy Bypass` part safely allows this one script to run without changing any system settings.

A2. Batch-file installer (simplest, double-click)

If you just want to **check** whether `fpalc` is installed and get guided instructions if it isn't:

1. **Double-click** `install_chromaprint.bat` in File Explorer.
2. Follow the on-screen messages.

The `.bat` file focuses on detection + clear instructions. For a fully automatic download + install, use the PowerShell installer (A1) above.

Option B: Manual installation

Step 1 — Download Chromaprint

1. Go to the official download page: <https://acoustid.org/chromaprint> (<https://acoustid.org/chromaprint>) (or the GitHub releases page: <https://github.com/acoustid/chromaprint/releases> (<https://github.com/acoustid/chromaprint/releases>)).
2. Under the latest release, download the **Windows** file. It is named like:

```
chromaprint-fpalc-1.6.0-windows-x86_64.zip
```

(The version number, e.g. `1.6.0`, may be newer — that's fine.)

Direct link for the current version:

https://github.com/acoustid/chromaprint/releases/download/v1.6.0/chromaprint-fpalc-1.6.0-windows-x86_64.zip (https://github.com/acoustid/chromaprint/releases/download/v1.6.0/chromaprint-fpalc-1.6.0-windows-x86_64.zip)

Step 2 — Extract the ZIP

1. In File Explorer, find the downloaded `.zip` file (usually in **Downloads**).
2. **Right-click** it → **Extract All...** → **Extract**.

3. Open the extracted folder. Inside you will find `fpcalc.exe` (it may be inside a sub-folder named like `chromaprint-fpcalc-1.6.0-windows-x86_64`).

Step 3 — Put `fpcalc.exe` somewhere the project can find it

You have two choices. **Either one works** — you do **not** need both.

Choice 1 (easiest): copy it into the project folder

1. Copy `fpcalc.exe`.
2. Paste it directly into this project folder (the one containing `acoustic_fingerprint.py` and this file).

That's it. When you run the tools from this folder, they will find `fpcalc.exe` right next to them.

Tip: If you put it somewhere else, you can tell the project exactly where it is by editing `config.json` and setting the full path, e.g.:

```
json
"acoustic": {
  "fpcalc_path": "C:\\Tools\\chromaprint\\fpcalc.exe"
}
```

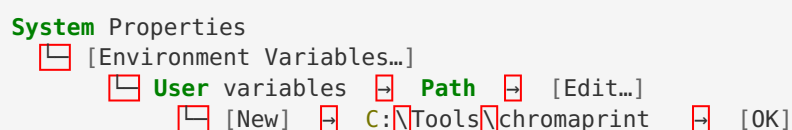
(Use double backslashes `\\` in JSON.)

Choice 2 (recommended for repeated use): add it to your PATH

Adding the folder to your **PATH** means any program — including Python — can find `fpcalc.exe` no matter which folder you're in.

1. Decide on a permanent home for it, e.g. create a folder `C:\Tools\chromaprint` and move `fpcalc.exe` there.
2. Press the **Windows key**, type **"environment variables"**, and click **"Edit the system environment variables"**.
3. In the window that opens, click the **"Environment Variables..."** button (bottom-right).
4. In the **top** box ("User variables for your name"), click the row named **Path**, then click **"Edit..."**.
 - If there is no **Path** row, click **"New..."** and create one named **Path**.
5. Click **"New"** and paste the folder path (e.g. `C:\Tools\chromaprint`).
 - **Important:** paste the folder, **not** the full path to `fpcalc.exe`.
6. Click **OK** on every window to save.
7. **Close and reopen** any PowerShell / Command Prompt windows so they pick up the new PATH.

Visual summary of the PATH dialog:



```
System Properties
[Environment Variables]
  [User variables] [Path] [Edit...]
    [New] [C:\Tools\chromaprint] [OK]
```

Verifying the installation

Open a **new** PowerShell or Command Prompt window and run:

```
fpcalc -version
```

You should see something like:

```
fpcalc version 1.6.0
```

If you copied `fpcalc.exe` **into the project folder** (Choice 1) but did **not** add it to PATH, test it from inside the project folder like this:

```
.\fpcalc.exe -version
```

Then confirm the project itself can use it. From the project folder run a quick acoustic identify (replace the file name with any audio/video clip you have):

```
python create_acoustic_fingerprint.py --file "some_clip.mp4" --identify
```

If you get past the “fpcalc not found” stage, you’re all set. 🎉

Troubleshooting

“fpcalc is not recognized as an internal or external command”

- You opened the terminal **before** adding it to PATH. Close the window and open a **new** one.
- Or you added the wrong thing to PATH. Make sure you added the **folder** (e.g. `C:\Tools\chromaprint`), **not** the file `fpcalc.exe` itself.
- Quick alternative: copy `fpcalc.exe` directly into the project folder (Choice 1 above) and run the project tools from there.

“running scripts is disabled on this system” (PowerShell)

- Run the installer with the bypass flag, which only affects that one run:

```
powershell
```

```
powershell -ExecutionPolicy Bypass -File .\install_chromaprint.ps1
```

Windows SmartScreen / antivirus warns about the download

- `fpcalc.exe` is a well-known open-source tool, but fresh downloads can trigger a generic warning. Download only from the official sources:
- <https://acoustid.org/chromaprint> (<https://acoustid.org/chromaprint>)
- <https://github.com/acoustid/chromaprint/releases> (<https://github.com/acoustid/chromaprint/releases>)
- On a SmartScreen prompt, click “**More info**” → “**Run anyway**” if you trust the official source.

The project still says it can't find `fpcalc`

- Tell it the exact location in `config.json` :

```
json
{
  "acoustic": {
    "fpcalc_path": "C:\\Tools\\chromaprint\\fpcalc.exe"
  }
}
```

- Or set an environment variable that `pyacoustid` also understands:

```
powershell
setx FPCALC "C:\\Tools\\chromaprint\\fpcalc.exe"
```

(Then open a new terminal.)

I downloaded the wrong file

- For 64-bit Windows (almost everyone today), use the file ending in `windows-x86_64.zip`. There is also a 32-bit `windows-i686.zip` for very old systems — only use that if your Windows is 32-bit.

ffmpeg errors when fingerprinting video files

- `fpcalc` handles plain audio on its own, but this project also uses **FFmpeg** to slice audio out of video. If you plan to fingerprint video files, install FFmpeg too: <https://www.gyan.dev/ffmpeg/builds/> (<https://www.gyan.dev/ffmpeg/builds/>) (download the “release” build, extract, and add its `bin` folder to PATH the same way as above).

Summary checklist

- [] Downloaded `chromaprint-fpalc-...-windows-x86_64.zip`
- [] Extracted `fpcalc.exe`
- [] Either copied it into the project folder **or** added its folder to PATH
- [] Opened a **new** terminal and `fpcalc -version` prints a version number
- [] (Optional) Installed FFmpeg if you want to fingerprint **video** files

You're ready to use the acoustic features:

```
python identify_audio.py --both           # live mic, phonetic + acoustic
python fingerprint_audio.py --file clip.mp4 --acoustic --identify
```